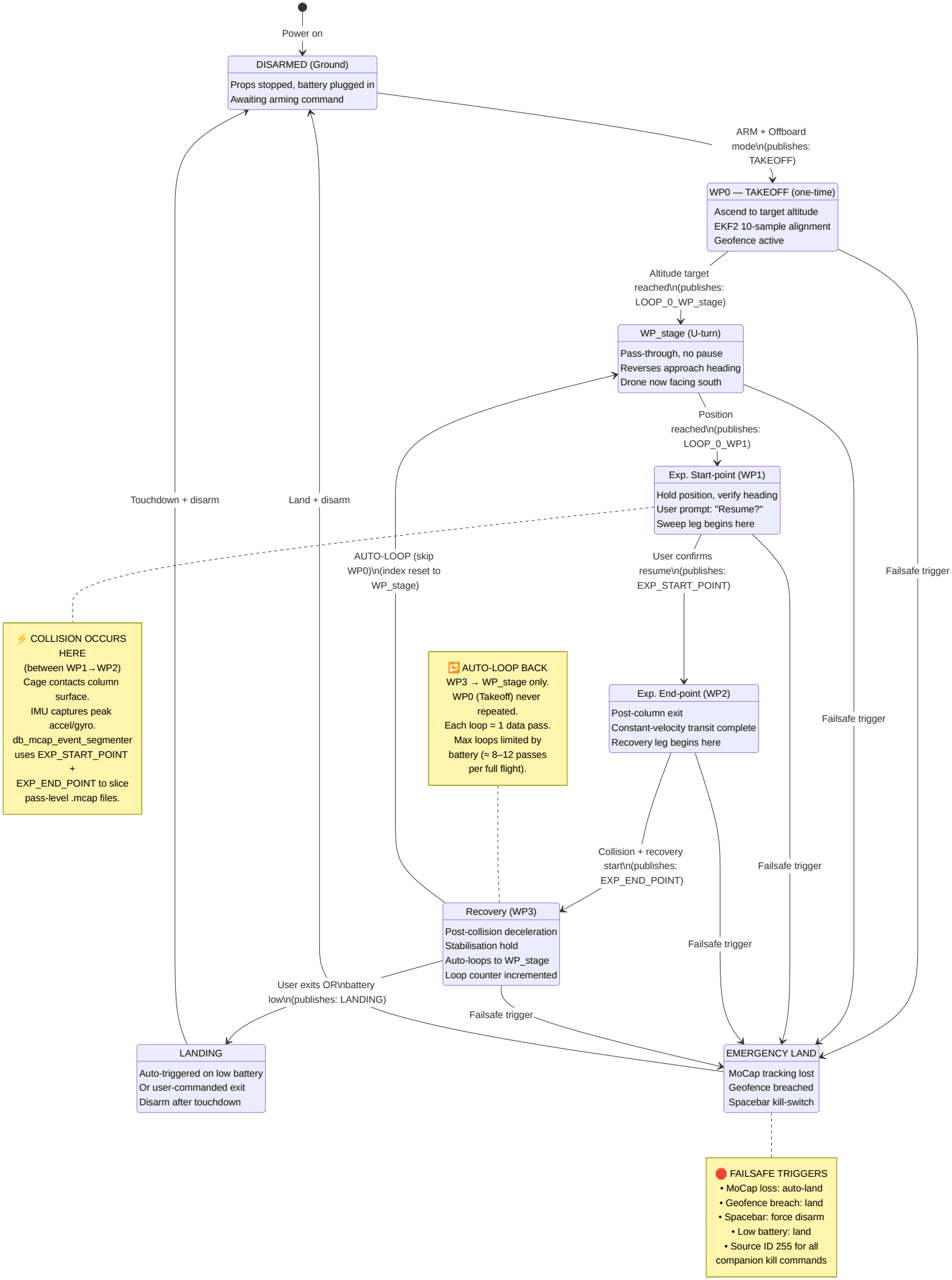
Power on
DISARMED (Ground)
Props stopped, battery plugged in
Awaiting arming command
ARM + Offboard mode\n(publishes: TAKEOFF)
WP0 — TAKEOFF (one-time)
Ascend to target altitude
EKF2 10-sample alignment
Geofence active
Altitude target reached\n(publishes: LOOP_0_WP_stage)
WP_stage (U-turn)
Pass-through, no pause
Reverses approach heading
Drone now facing south
Position reached\n(publishes: LOOP_0_WP1)
Exp. Start-point (WP1)
Hold position, verify heading
User prompt: "Resume?"
Sweep leg begins here
User confirms resume\n(publishes: EXP_START_POINT)
Exp. End-point (WP2)
Post-column exit
Constant-velocity transit complete
Recovery leg begins here
Collision + recovery start\n(publishes: EXP_END_POINT)
Recovery (WP3)
Post-collision deceleration
Stabilisation hold
Auto-loops to WP_stage
Loop counter incremented
User exits OR\nbattery low\n(publishes: LANDING)
LANDING
Auto-triggered on low battery
Or user-commanded exit
Disarm after touchdown
EMERGENCY LAND
MoCap tracking lost
Geofence breached
Spacebar kill-switch
Touchdown + disarm
Land + disarm
AUTO-LOOP (skip WP0)\n(index reset to WP_stage)
Failsafe trigger
Failsafe trigger
Failsafe trigger
Failsafe trigger
Failsafe trigger
⚡ COLLISION OCCURS HERE
(between WP1→WP2)
Cage contacts column surface.
IMU captures peak accel/gyro.
db_mcap_event_segmenter uses EXP_START_POINT + EXP_END_POINT to slice pass-level .mcap files.
🔁 AUTO-LOOP BACK
WP3 → WP_stage only.
WP0 (Takeoff) never repeated.
Each loop = 1 data pass.
Max loops limited by battery (≈ 8–12 passes per full flight).
🛑 FAILSAFE TRIGGERS
• MoCap loss: auto-land
• Geofence breach: land
• Spacebar: force disarm
• Low battery: land
• Source ID 255 for all companion kill commands